TEMASEK POLYTECHNIC
SCHOOL OF INFORMATICS & IT

DIPLOMA IN APPLIED ARTICIFICAL INTELLIGENCE
DIPLOMA IN BIG DATA & ANALYTICS
COMMON ICT PROGRAMME
DIPLOMA IN CYBERSECURITY & DIGITAL FORENSICS
DIPLOMA IN IMMERSIVE MEDIA & GAME DEVELOPMENT
DIPLOMA IN INFORMATION TECHNOLOGY

AY2025/2026 OCTOBER SEMESTER

CASE STUDY (30%)

NETWORK AND CLOUD TECHNOLOGY (CCF1C04)

SUBJECT LEVEL: 1

INSTRUCTIONS TO CANDIDATES

1. This Case Study consists of **8** pages (including cover page and appendix).
2. The objective is to assess students in the following key areas:
   a. Understanding the basic Cloud Services concepts
   b. Application of the Cloud Services concepts
3. You are to form a group consists of minimum of 2 and up to 4 members and divide the proposed tasks among your group members. Each member is required to complete their allocated tasks for the Case Study individually.
4. One report per group is required for submission.

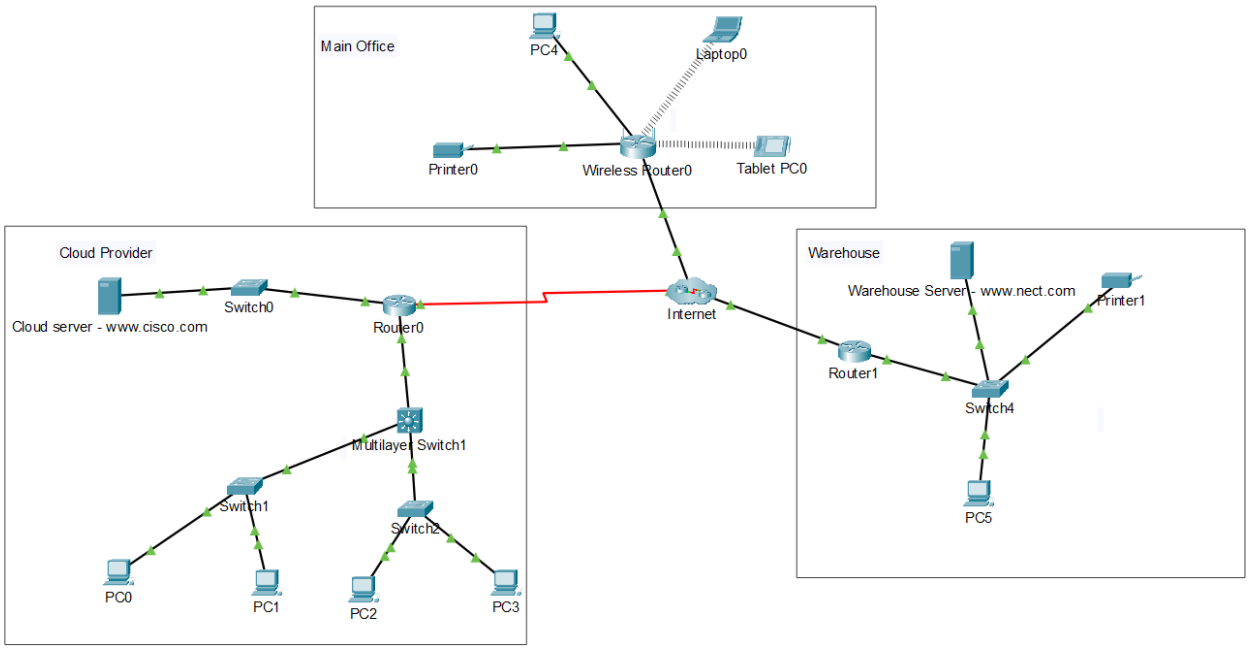
Scenario

NECT Pte Ltd is a traditional SME logistic company based in Singapore. NECT Pte Ltd has been operating for the past 15 years. The company specialises in last-mile delivery, warehouse storage, and distribution for e-commerce and small retailers. The company's network has expended incrementally without a clear long-term plan. The company background information is summarised below:

Company Information	**Details**
Employees	150 staff with the following breakdown: • Admin & Finance: 15 • Warehouse Operations: 60 • Delivery Drivers/Riders: 40 (Full-time) and 22 (Part-time) • Customer Service: 10 • IT Team: 3 (mostly junior roles)
Annual Revenue	Around SGD 8M (based on previous Financial Year performance)
Physical Footprint	1 small HQ and 1 industrial warehouse (in Ubi area)
Key Characteristics	• Stable business, but not digitally modern • IT systems expanded incrementally without a clear long-term plan • Operations heavily reliant on staff to be physically onsite
IT Services	• Internet access • Basic cloud-based solutions are used o A basic static corporate web site hosted on Wix that provides contact information o Gmail for email service
Applications running On Premise	• Email Server (ES) • Web Server (WS) • Warehouse Management System (WMS); • Transportation Management System (TMS) • Fleet Management System (FMS) • Order Management System (OMS) • File Sharing Server (FS) • Print Server (PS) • Basic database server for orders/inventory

Company Information	Details
Operations with on-premise systems	• All systems (ES/WS/WMS/TMS/FMS/OMS/FS/PS) run on local servers • Warehouse operations require LAN, scanners and local WMS • Delivery drivers use manual job sheets • Customer service staff updates orders manually • Staff must be in the office because systems do not support remote access • Admin staff has limited IT knowledge, causing slow updates and high risks • Entire business depends mostly on hardware inside the office

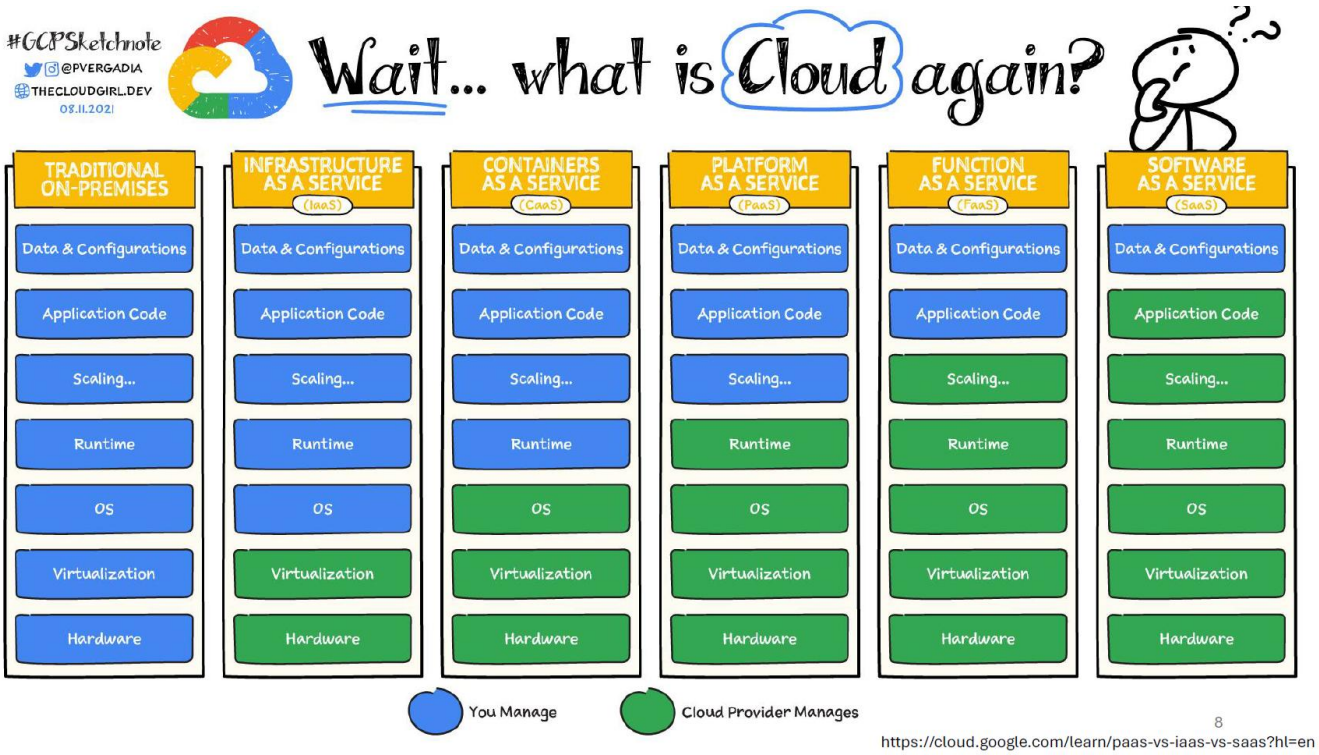
Current company network topology as shown below.



Tasks

You are hired as a Junior Network Technicians to help the company in improving their operation. The company is exploring the feasibility of migrating to a cloud deployment for either some or all their existing applications (i.e. ES/WS/WMS/TMS/FMS/OMS/FS/PS) currently running on local servers.

You are required to complete the following tasks using AI as your personal tutor.



1. As a group, choose one Cloud Service from the Cloud Services listed above for your research. Each member selects and concentrates on two distinct key areas from the list provided below:

   i. Availability
   ii. Performance
   iii. Compliance and Governance
   iv. Scalability
   v. Cost
   vi. Security
   vii. Features
   viii. Sustainability
   ix. Operation
   x. Vendor lock-in risk

2. Your research should cover the following: -

a. Comparison of your chosen Cloud Service with the other Cloud Services
b. Explain how your chosen Cloud Service is applicable to NECT's current on-premise setup
c. Provide a comprehensive assessment on the pros and cons of chosen Cloud Service, and clearly justify why the recommended cloud service aligns best with the company's requirements

3. Evaluate AI's responses critically for accuracy and quality, then enhance it using your own knowledge and judgement. Illustrates your collaboration with AI by:
    a. Specifying which task was AI used for
    b. Recording AI's generated output
    c. Documenting your critiques
    d. Documenting your improvements

Submission

You are required to submit **One** group report, not more than 40 pages (excluding cover page, table of content and appendixes), via POLITEMall. Refer to the Teaching Plan for the submission deadline. You may use the report template in Appendix A.

Assessment Scheme

Your report will be assessed on the following criteria.

- Research topic (15%)
- Report writing (15%)

Note: Plagiarism is not allowed and will result in a zero for the assignment. You are required to cite any sources used in your Case Study.

Late submissions

Late submissions will be imposed as follows.

- Late and < 1 day – 10% deduction from absolute marks given for the part of the work e.g. if the assignment was worth 100 marks, you were given 75 marks for the work, after the penalty you are left with 65 marks i.e. 75 – 100x10%
- Late >= 1 and < 2 days – 20% deduction from absolute marks
- Late >= 2 days – no marks will be awarded

~~ End of Case Study ~~

Appendix A - Report Template

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Diploma in [ ]

Case Study Report

Student Name (Matric Number) :

Tutorial Group :

Tutor :

Submission Date :

Declaration of Originality

I am the originator of this work and I have appropriately acknowledged all other original sources used as my references for this work.

I understand that Plagiarism is the act of taking and using the whole or any part of another person's work, including work generated by AI, and presenting it as my own.

I understand that Plagiarism is an academic offence

and if I am found to have committed or abetted the offence of plagiarism in relation to this submitted work, disciplinary action will be enforced.

Table of Content

1. Introduction

2. Research
   2.1 Research #1 (Member #1)
      2.1.1 Comparison
      2.1.2 Pros and Cons
      2.1.3 Justification
      2.1.4 Document of collaboration with AI
   2.2 Research #2 (Member #2)
   2.3 …

3. Appendixes
   A- References
   B- …
   C- …